HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Slides to the lecture

Chapter 1: Basics

Prof. Dr. Markus Pfeil

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1 Basics

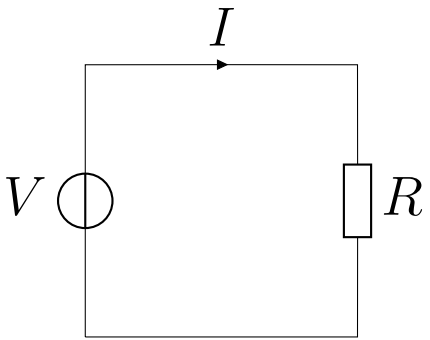
1.1 Quantities, Units and Ohm's Law

Quantities and units

quantity	symbol	unit	definition
voltage	V	volt, V	energy per charge, $\mathrm{J/C}$
current	I	ampere, A	charge per time, $\mathrm{C/s}$
resistance	R	ohm, Ω	$R = V/I$
power	P	watt, W	$P = V \cdot I$
capacitance	C	farad, F	$C = Q/V$
time, frequency	t, f	s, Hz	$f = 1/T$

- prefixes: p (10^{-12}), n (10^{-9}), µ (10^{-6}), m (10^{-3}), k (10^{3}), M (10^{6})
- useful combination: V / kΩ = mA, mA · kΩ = V

Ohm's law



Ohm's law

$$V = R \cdot I \qquad I = \frac{V}{R} \qquad R = \frac{V}{I}$$

- a resistor is **linear**: double voltage $\Rightarrow$ double current
- the conductance $G = 1/R$ is measured in siemens, S

Power and power rating

Power in a resistor

$$P = V \cdot I = I^2 R = \frac{V^2}{R}$$

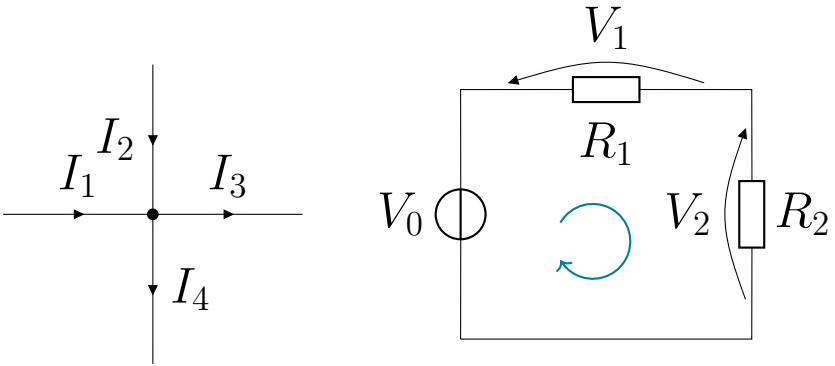
- the power is turned into heat; every resistor has a **power rating**
  - SMD 0603: 0.1 W, SMD 1206: 0.25 W, through-hole (0207): 0.25 W … 0.6 W
- design rule: stay below about 50 % of the rating
- small signal circuits (kΩ, mA) dissipate only mW
- example: $100\,\Omega$ at 5 V: $P = 25\,\text{V}^2/100\,\Omega = 250$ mW, too much for a 0.25 W resistor

Standard values: the E-series

- resistors are not available in every value, but in logarithmic series
- E12 ($10\,\%$ tolerance): 1.0 1.2 1.5 1.8 2.2 2.7 3.3 3.9 4.7 5.6 6.8 8.2
- E24 ($5\,\%$) adds: 1.1 1.3 1.6 2.0 2.4 3.0 3.6 4.3 5.1 6.2 7.5 9.1
- E96 ($1\,\%$): 96 values per decade, standard for metal film resistors today
- each value exists in every decade: $4.7\,\Omega$, $47\,\Omega$, ..., $4.7\,\mathrm{M}\Omega$

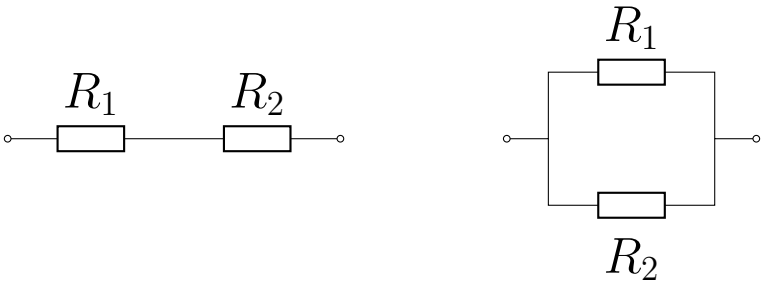
1.2 Kirchhoff's Laws and Networks

Kirchhoff's current law (KCL) and voltage law (KVL)



- KCL: at every node the sum of incoming currents equals the sum of outgoing currents: $I_1 + I_2 = I_3 + I_4$
- KVL: around every closed loop the sum of voltages is zero: $V_0 = V_1 + V_2$

Series and parallel connection

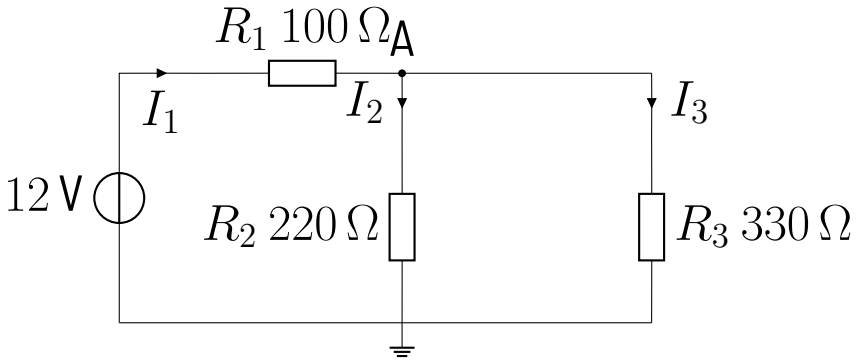


Series and parallel

series: $R = R_1 + R_2$ parallel: $R = R_1 \parallel R_2 = \dfrac{R_1 R_2}{R_1 + R_2}$

- series: same current, the voltages add up
- parallel: same voltage, the currents add up
- parallel resistance is always smaller than the smallest resistor; $R \parallel R = R/2$

Example: series-parallel network



- $R_2 \parallel R_3 = 132\,\Omega$, total $R = 232\,\Omega$, $I_1 = 12\,\text{V}/232\,\Omega = 51.7\,\text{mA}$
- $V_A = I_1 \cdot 132\,\Omega = 6.83\,\text{V}$; $I_2 = 31.0\,\text{mA}$, $I_3 = 20.7\,\text{mA}$ (KCL: $I_1 = I_2 + I_3$)

▶ Falstad: `examples/basics/network.txt`

Exercise 1.1: Power in a series-parallel network

Use the network above ($12\,\mathrm{V}$, $R_1 = 100\,\Omega$, $R_2 = 220\,\Omega$, $R_3 = 330\,\Omega$). All resistors are $0.25\,\mathrm{W}$ types.

a) Calculate the power dissipated in each resistor and the power delivered by the source.
b) Which resistor is overloaded? Which rating would you choose instead?
c) Check the currents and powers in Falstad.

Solution 1.1

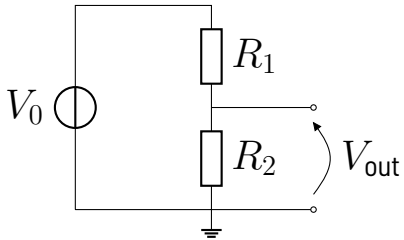
a) $P_1 = I_1^2 R_1 = (51.7\,\text{mA})^2 \cdot 100\,\Omega = 268\,\text{mW}$, $P_2 = V_A^2/R_2 = (6.83\,\text{V})^2/220\,\Omega = 212\,\text{mW}$, $P_3 = (6.83\,\text{V})^2/330\,\Omega = 141\,\text{mW}$. Source: $P = 12\,\text{V} \cdot 51.7\,\text{mA} = 621\,\text{mW} = P_1 + P_2 + P_3$.

b) R_1 exceeds $0.25\,\text{W}$; R_2 is at $85\,\%$ of its rating, which is too hot as well. With the $50\,\%$ rule R_1 needs $0.6\,\text{W}$ and R_2 $0.5\,\text{W}$ (better: $1\,\text{W}$ for R_1).

c) ▶ Falstad: `examples/basics/network.txt`: `VA` $= 6.83\,\text{V}$; hovering over the resistors shows $I_1 = 51.7\,\text{mA}$, $I_2 = 31.0\,\text{mA}$, $I_3 = 20.7\,\text{mA}$ and the power of each resistor.

1.3 The Voltage Divider

Unloaded divider



- the same current flows through R_1 and R_2: $I = V_0/(R_1 + R_2)$
- the voltages divide in the ratio of the resistances

Voltage divider

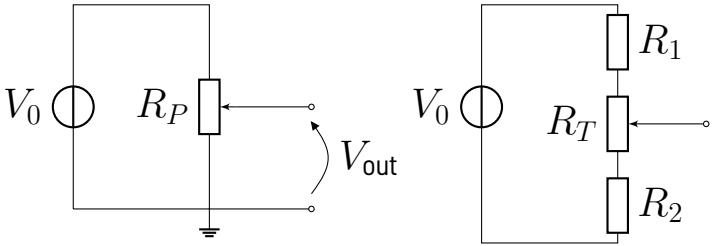
$$V_{\text{out}} = V_0 \frac{R_2}{R_1 + R_2}$$

Loaded divider

- a load R_L appears in parallel to R_2: $V_{\text{out}} = V_0 \dfrac{R_2 \parallel R_L}{R_1 + R_2 \parallel R_L}$
- seen from the load the divider is a real source with $V_{\text{th}} = V_0\, R_2/(R_1 + R_2)$ and $R_{\text{th}} = R_1 \parallel R_2$
- rule of thumb: $R_L \geq 10\,(R_1 \parallel R_2)$ keeps the error below $10\,\%$
- for an error below $1\,\%$: $R_L \geq 100\,(R_1 \parallel R_2)$, or buffer the tap with an op-amp

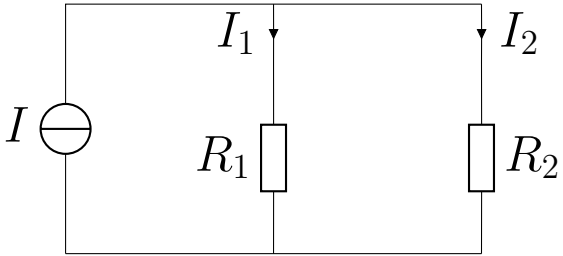
▶ Falstad: `examples/basics/divider_loaded.txt`

Potentiometer: adjustable divider



- the wiper divides the track: $V_{\text{out}} = V_0 \cdot x, x = 0 \ldots 1$ (linear taper)
- trimmer between two fixed resistors: fine adjustment over a small range
- the wiper must not carry a large current; load it lightly

Current divider



Current divider

$$I_1 = I \frac{R_2}{R_1 + R_2} \qquad I_2 = I \frac{R_1}{R_1 + R_2}$$

- the **smaller** resistor takes the **larger** current
- example: 10 mA into $1\,\text{k}\Omega \,||\, 4.7\,\text{k}\Omega$: $I_1 = 8.25$ mA, $I_2 = 1.75$ mA

▶ Falstad: `examples/basics/current_divider.txt`

Exercise 1.2: Loaded voltage divider

A divider with $R_1 = R_2 = 10\,\text{k}\Omega$ is supplied with $V_0 = 10\,\text{V}$.

a) Calculate V_{out} without load.
b) Calculate V_{out} with a load $R_L = 10\,\text{k}\Omega$.
c) Check both results in Falstad.

Solution 1.2

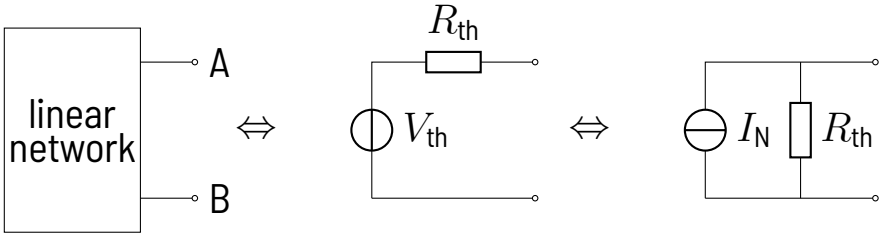
a) $V_{\text{out}} = 10\,\text{V} \cdot \frac{10}{10+10} = 5\,\text{V}$.

b) $R_2 \parallel R_L = 5\,\text{k}\Omega$, so $V_{\text{out}} = 10\,\text{V} \cdot \frac{5}{10+5} = 3.33\,\text{V}$. The load costs one third of the output voltage, because R_L is only twice the source resistance $R_{\text{th}} = 5\,\text{k}\Omega$.

c) ▶ Falstad: `examples/basics/divider_loaded.txt`
the labels show `Vout_open` $= 5.00\,\text{V}$ and `Vout_load` $= 3.33\,\text{V}$ (move the mouse over the label).

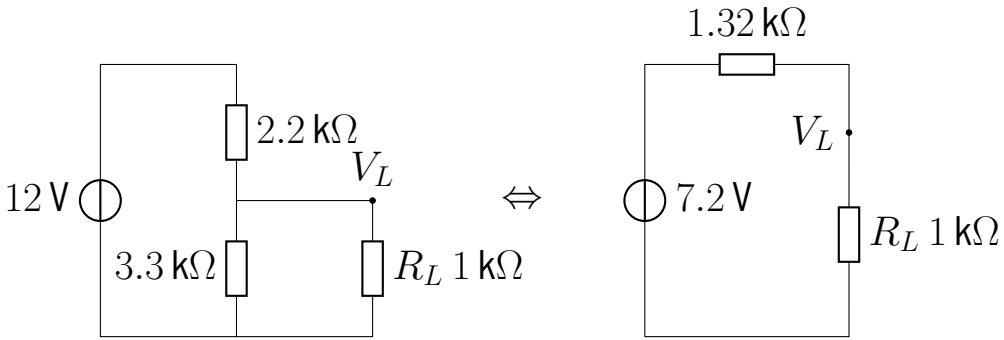
1.4 Equivalent Sources and Superposition

Thevenin and Norton equivalent



- open-circuit voltage: $V_{\text{th}} = V_{\text{oc}}$ (nothing connected to A–B)
- short-circuit current: $I_{\text{N}} = I_{\text{sc}}$ (A and B shorted)
- internal resistance $R_{\text{th}} = V_{\text{oc}}/I_{\text{sc}}$, or: set all sources to zero (voltage source $\rightarrow$ wire, current source $\rightarrow$ open) and calculate the resistance between A and B

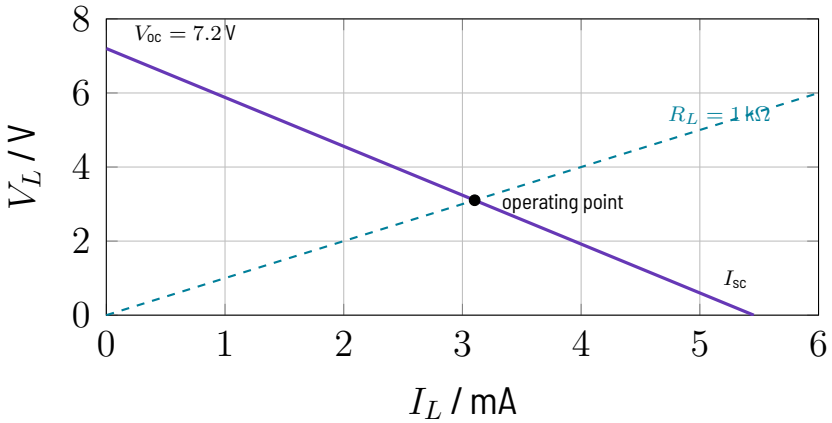
Example: Thevenin equivalent of a divider



- $V_{\text{oc}} = 12\,\text{V} \cdot 3.3/5.5 = 7.2\,\text{V}, \quad I_{\text{sc}} = 12\,\text{V}/2.2\,\text{k}\Omega = 5.45\,\text{mA}$
- $R_{\text{th}} = 7.2\,\text{V}/5.45\,\text{mA} = 1.32\,\text{k}\Omega = 2.2\,\text{k}\Omega \parallel 3.3\,\text{k}\Omega$
- with R_L: $V_L = 7.2\,\text{V} \cdot 1/(1 + 1.32) = 3.10\,\text{V}$ in both circuits

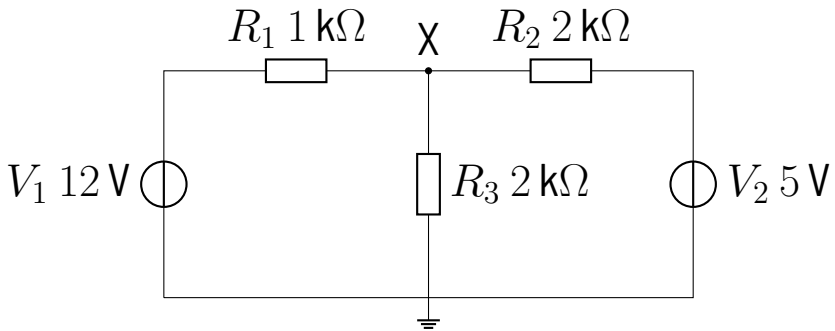
▶ Falstad: `examples/basics/thevenin.txt`

Load line of a real source



- the source can deliver every point on the line $V_L = V_{th} - R_{th}\, I_L$
- the load forces $V_L = R_L\, I_L$; the intersection is the operating point

Superposition



- in a linear network the effect of all sources is the sum of the effects of each source alone
- the other sources are set to zero: voltage source $\rightarrow$ wire, current source $\rightarrow$ open
- V_1 alone: $V_{X1} = 12\,\text{V} \cdot \frac{R_2 \parallel R_3}{R_1 + R_2 \parallel R_3} = 6\,\text{V}$; V_2 alone: $V_{X2} = 5\,\text{V} \cdot \frac{R_1 \parallel R_3}{R_2 + R_1 \parallel R_3} = 1.25\,\text{V}$
- both: $V_X = 7.25\,\text{V}$ ▶ Falstad: `examples/basics/superposition.txt`

Exercise 1.3: Thevenin equivalent from open-circuit voltage and short-circuit current

The divider $12\,\mathrm{V}$, $R_1 = 2.2\,\mathrm{k\Omega}$ (top), $R_2 = 3.3\,\mathrm{k\Omega}$ (bottom) feeds a load $R_L = 1\,\mathrm{k\Omega}$.

a) Determine V_{oc} and I_{sc} at the tap and from them R_{th}.
b) Calculate V_L with the equivalent circuit.
c) Which load would reduce the open-circuit voltage by only $5\,\%$?
d) Compare the original circuit and the equivalent in Falstad.

Solution 1.3

a) $V_{\text{oc}} = 12\,\text{V} \cdot 3.3/(2.2+3.3) = 7.2\,\text{V}$. With the tap shorted no current flows in R_2: $I_{\text{sc}} = 12\,\text{V}/2.2\,\text{k}\Omega = 5.45\,\text{mA}$. $R_{\text{th}} = 7.2\,\text{V}/5.45\,\text{mA} = 1.32\,\text{k}\Omega$.

b) $V_L = 7.2\,\text{V} \cdot \frac{1}{1+1.32} = 3.10\,\text{V}$.

c) $V_L/V_{\text{oc}} = R_L/(R_L + R_{\text{th}}) = 0.95 \Rightarrow R_L = 19\,R_{\text{th}} = 25.1\,\text{k}\Omega$ (E12: $27\,\text{k}\Omega$).

d) ▶ Falstad: `examples/basics/thevenin.txt`: `VL_orig` = `VL_equiv` = $3.10\,\text{V}$. Replace R_L by a wire and hover over it: the short-circuit current is $5.45\,\text{mA}$ in both circuits.

Exercise 1.4: Superposition

Use the circuit of the superposition slide.

a) Calculate V_X with superposition.
b) Check the result with KCL at node X.
c) How large is the current through R_2, and in which direction does it flow?
d) Check the three cases in Falstad.

Solution 1.4

a) V_1 alone: $R_2 \parallel R_3 = 1\,\text{k}\Omega$, $V_{X1} = 12\,\text{V} \cdot 1/2 = 6\,\text{V}$. V_2 alone: $R_1 \parallel R_3 = 0.667\,\text{k}\Omega$, $V_{X2} = 5\,\text{V} \cdot 0.667/2.667 = 1.25\,\text{V}$. $V_X = 7.25\,\text{V}$.

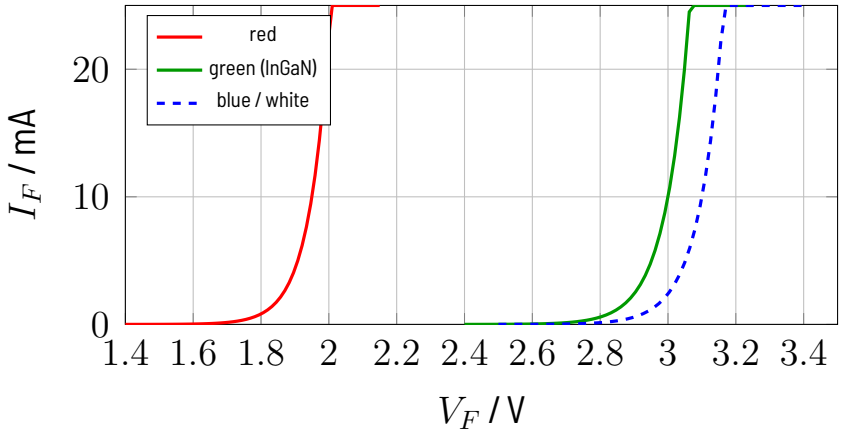
b) $\frac{12-7.25}{1} = 4.75\,\text{mA}$ flows in through R_1; $\frac{7.25}{2} = 3.625\,\text{mA}$ flows out through R_3 and $\frac{7.25-5}{2} = 1.125\,\text{mA}$ through R_2: $4.75 = 3.625 + 1.125$ ✓

c) $1.125\,\text{mA}$ from X into the $5\,\text{V}$ source: V_2 is charged (absorbs power).

d) ▶ Falstad: `examples/basics/superposition.txt`: `VX1` $= 6.00\,\text{V}$, `VX2` $= 1.25\,\text{V}$, `VX` $= 7.25\,\text{V}$.

1.5 The LED with Series Resistor

LED characteristic



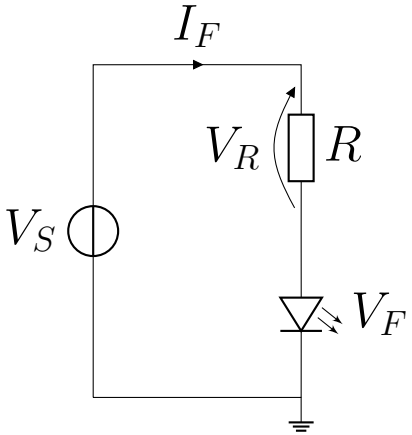
- below a threshold almost no current, above it the current rises steeply
- simple model: constant forward voltage V_F at the operating current

Forward voltage and current

colour	material	V_F at $10\,\text{mA} \ldots 20\,\text{mA}$
infrared	GaAs	$1.2\,\text{V} \ldots 1.5\,\text{V}$
red	AlGaInP	$1.8\,\text{V} \ldots 2.1\,\text{V}$
yellow, orange	AlGaInP	$2.0\,\text{V} \ldots 2.2\,\text{V}$
green (classic, yellow-green)	GaP	$2.1\,\text{V} \ldots 2.4\,\text{V}$
green (true green), blue, white	InGaN	$2.8\,\text{V} \ldots 3.4\,\text{V}$

- standard $3\,\text{mm}$/$5\,\text{mm}$ LEDs: $I_F \leq 20\,\text{mA}$; modern LEDs are clearly visible at $2\,\text{mA} \ldots 5\,\text{mA}$
- reverse voltage only about $5\,\text{V}$: an LED is not a rectifier

Dimensioning the series resistor



LED series resistor

$$R = \frac{V_S - V_F}{I_F} \qquad P_R = (V_S - V_F) \cdot I_F$$

- example: $V_S = 5\,\text{V}$, red LED $V_F = 2.0\,\text{V}$, $I_F = 10\,\text{mA}$: $R = 300\,\Omega \Rightarrow$ E12: $330\,\Omega$, $I_F = 9.1\,\text{mA}$, $P_R = 27\,\text{mW}$

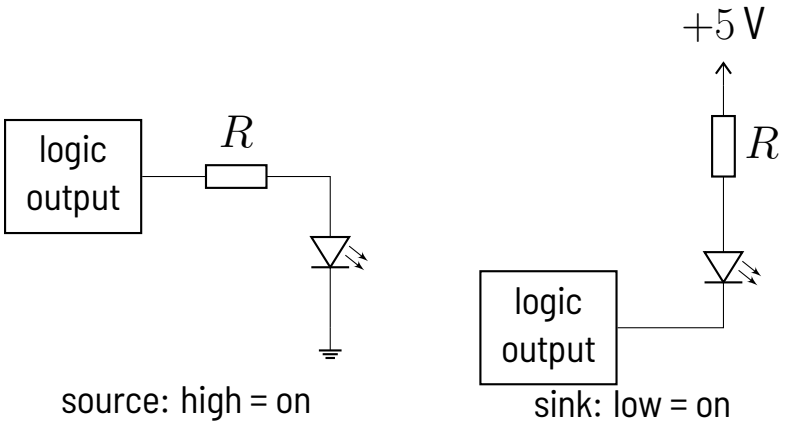
Design recipe

Design card: LED with series resistor

1. take V_F from the datasheet at the desired current (or the table: red 2.0 V, blue/white 3.0 V)
2. choose I_F: 2 mA … 10 mA for an indicator, check the maximum current of the driver
3. $R = (V_S - V_F)/I_F$, round **up** to the next E12/E24 value
4. recalculate I_F and check $P_R = I_F^2 R$ against the rating

- rounding up keeps the current below the target, which protects LED and driver
- the larger $V_S - V_F$, the less I_F depends on the spread of V_F

Driving an LED from a logic output



- a logic output is a switch to V_S or to ground with an output resistance of some $10\,\Omega$
- 74HC at 5 V: ± 4 mA with guaranteed levels, absolute maximum 25 mA per pin
- microcontrollers: some mA per pin and a limit for the whole port
- $330\,\Omega$ at 5 V gives about 9 mA, 1 kΩ about 3 mA

▶ Falstad: `examples/basics/led_resistor.txt`

Exercise 1.5: LED on a $5\,\mathrm{V}$ logic output

A red LED ($V_F = 2.0\,\mathrm{V}$ at $10\,\mathrm{mA}$) is to be driven by a $5\,\mathrm{V}$ logic output.

a) Calculate the series resistor for $10\,\mathrm{mA}$ and choose an E12 value.
b) Calculate the actual current and the power in the resistor.
c) The same LED is to be driven from a $3.3\,\mathrm{V}$ microcontroller pin with at most $4\,\mathrm{mA}$. Choose the resistor.
d) Falstad uses an LED model with $V_F \approx 1.8\,\mathrm{V}$. Which current does the simulation show with $330\,\Omega$?

Solution 1.5

a) $R = (5\,\mathrm{V} - 2.0\,\mathrm{V})/10\,\mathrm{mA} = 300\,\Omega$; rounded up to E12: $330\,\Omega$.

b) $I_F = 3.0\,\mathrm{V}/330\,\Omega = 9.1\,\mathrm{mA}$, $P_R = 3.0\,\mathrm{V} \cdot 9.1\,\mathrm{mA} = 27\,\mathrm{mW}$: any resistor will do, even SMD 0402.

c) $R \geq (3.3\,\mathrm{V} - 2.0\,\mathrm{V})/4\,\mathrm{mA} = 325\,\Omega$, again $330\,\Omega$ ($I_F = 3.9\,\mathrm{mA}$; V_F is a bit lower at this current, so the real current is slightly higher).

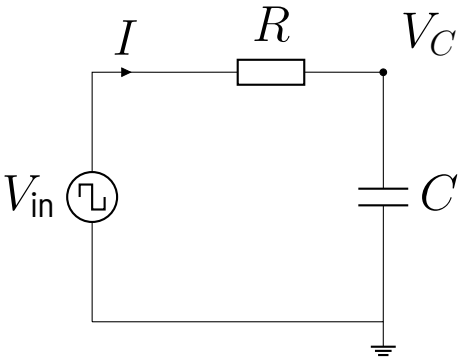
d) ▶ Falstad: `examples/basics/led_resistor.txt`: $\mathtt{VLED} = 1.78\,\mathrm{V}$, so $I_F = (5 - 1.78)\,\mathrm{V}/330\,\Omega = 9.76\,\mathrm{mA}$ (hover over the resistor). Use the slider to see how the current changes with R.

1.6 Capacitor and RC Circuits

The capacitor

- charge proportional to voltage: $Q = C \cdot V$
- current = change of charge: $I = C \dfrac{dV}{dt}$
  - the voltage across a capacitor cannot jump (that would need an infinite current)
  - DC: once charged, no current flows (open circuit)
- energy: $E = \frac{1}{2} C V^2$
- types: ceramic (pF to 100 µF; COG stable, X7R voltage dependent), film (precise, for filters and hold capacitors), electrolytic (large, polarized)

Charging through a resistor

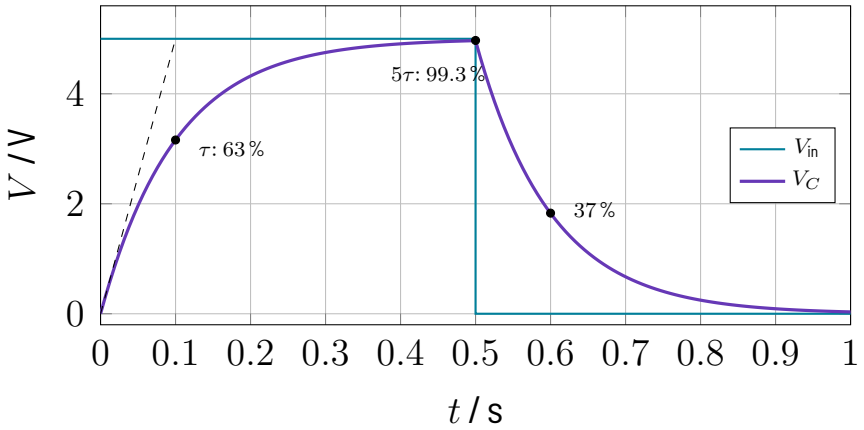


RC charging and discharging

$$V_C(t) = V_0\left(1 - e^{-t/\tau}\right) \qquad V_C(t) = V_0\, e^{-t/\tau} \qquad \tau = R\,C$$

- step of V_0 at the input; the current $I = (V_{\text{in}} - V_C)/R$ decreases as V_C rises
- time to reach a voltage V_1: $t = \tau \ \ln \frac{V_0}{V_0 - V_1}$

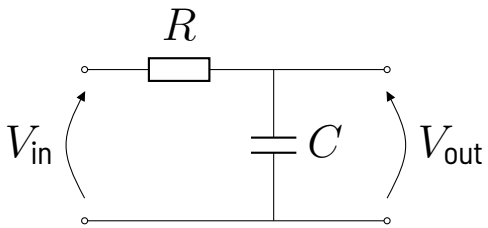
The time constant

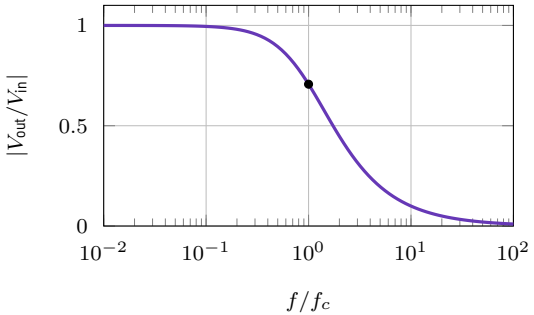


- after τ: $63\,\%$, after 3τ: $95\,\%$, after 5τ: $99.3\,\%$ (“settled”)
- the tangent at $t = 0$ reaches the final value after exactly τ
- example $R = 10\,\text{k}\Omega$, $C = 10\,\mu\text{F}$: $\tau = 0.1\,\text{s}$

▶ Falstad: `examples/basics/rc_charging.txt`

Preview: the RC low-pass





Cut-off frequency

$$f_c = \frac{1}{2\pi RC} \qquad \left|\frac{V_{\text{out}}}{V_{\text{in}}}\right| = \frac{1}{\sqrt{1+(f/f_c)^2}}$$

- at f_c the amplitude drops to $1/\sqrt{2} = 0.707$ (-3 dB), phase $-45°$
- example: $1.6\,\text{k}\Omega$, $100\,\text{nF}$: $f_c = 995\,\text{Hz}$ ▶ Falstad: `examples/basics/rc_lowpass.txt`

Exercise 1.6: RC charging

A capacitor $C = 10\,\mu\text{F}$ is charged through $R = 10\,\text{k}\Omega$ from a square wave that switches between $0\,\text{V}$ and $5\,\text{V}$ at $1\,\text{Hz}$.

a) Calculate τ and V_C after τ and after $0.5\,\text{s}$.
b) After which time does V_C reach $4\,\text{V}$?
c) To which voltage does C discharge $0.1\,\text{s}$ after the falling edge?
d) Read the values from the Falstad scope.

Solution 1.6

a) $\tau = 10\,\text{k}\Omega \cdot 10\,\mu\text{F} = 0.1\,\text{s}$. $V_C(\tau) = 5\,\text{V}\,(1 - e^{-1}) = 3.16\,\text{V}$; $V_C(0.5\,\text{s}) = V_C(5\tau) = 5\,\text{V}\,(1 - e^{-5}) = 4.97\,\text{V}$.

b) $t = \tau \ln \frac{5}{5-4} = 0.1\,\text{s} \cdot \ln 5 = 0.161\,\text{s}$.

c) The discharge starts at $4.97\,\text{V}$: $V_C = 4.97\,\text{V} \cdot e^{-1} = 1.83\,\text{V}$.

d) ▶ Falstad: `examples/basics/rc_charging.txt`: the scope shows $3.16\,\text{V}$ at $0.1\,\text{s}$, $4\,\text{V}$ at $0.161\,\text{s}$, $4.97\,\text{V}$ at the falling edge and $1.83\,\text{V}$ $0.1\,\text{s}$ later.

1.7 Working with Falstad

Opening and building circuits

- `https://www.falstad.com/circuit/circuitjs.html`
- open an example: click its Falstad box in the script, or **File** → **Import From Text** and paste the content of `examples/basics/*.txt`
- new circuit: **Circuits** → **Blank Circuit**
- place elements: **Draw** menu or right click; frequent ones have shortcuts (`r` resistor, `c` capacitor, `w` wire, `g` ground)
- drag to place, double click to edit values (`4.7k`, `100n`, `10u` are accepted)
- save your work: **File** → **Export As Text** or **Export As Link**

Pitfalls when drawing

- wires connect **only at their end points**: a wire passing over a terminal is not connected; split the wire there
- unconnected terminals are shown as red dots
- every circuit needs a ground (or a source that defines the potential)
- **labeled nodes** (**Draw** $\rightarrow$ **Wiring** $\rightarrow$ **Labeled Node**) with the same name are connected without a wire, and their name can be used for scopes
- "singular matrix": usually a floating node or two ideal voltage sources in parallel

Measuring

- hover the mouse over an element: the bottom right corner shows voltage, current and power
- hover over a labeled node: its voltage to ground
- voltage colours: green positive, red negative, grey zero; yellow dots: current
- scope: right click on an element → **View in New Scope**; right click on a scope for settings (scale, several traces, **Max Scale**)
- sliders: **Edit** dialog of an element, check **Show slider**; the examples have sliders for the values worth changing
- simulation speed: slider on the right; the time step is in **Options → Other Options**